



Modelling Financial Markets

Seminar 12: Options and the Volatility Surface — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ ATM: At-The-Money
- ▣ CI: Confidence Interval
- ▣ DVOL: Deribit Volatility Index
- ▣ OTM: Out-of-The-Money
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ VIX: CBOE Volatility Index
- ▣ VIX3M: Cboe 3-Month Volatility Index
- ▣ VRP: Variance Risk Premium
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Detect a parity arbitrage and price options on a binomial tree and with Black–Scholes step by step
- Build delta and delta-gamma hedges and measure the hedging error with a confidence interval
- Fit a smile, compute a VIX-type index and test the variance risk premium with robust standard errors

- Background: Lecture 12; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- **Part A:** eight pen-and-paper exercises (A1–A8)
- **Part B:** seven computational exercises with inference (B1–B7)
- **Part C:** an open question, a seed for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Volatility in annualised %; “vol points” = percentage points of volatility; time in years unless stated
- Overlapping windows are autocorrelated: always use Newey–West standard errors or a block bootstrap [Newey & West \(1987\)](#), [Künsch \(1989\)](#)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (parity arbitrage), A3 (one-period binomial tree), A5 (Black–Scholes and the Greeks, step by step), A7 (implied volatility by Newton)
- Part B: B1 (hedging error with bootstrap intervals), B3 (SVI fit of a Bitcoin smile), B5 (variance risk premium of the S&P 500)

Proposed: you solve them

- Part A: A2 (parity with dividends; model: A1), A4 (American put; model: A3), A6 (delta-gamma hedge; model: A5), A8 (variance from an option strip; model: A7)
- Part B: B2 (delta-hedged S&P 500 options; model: B1, B5), B4 (a VIX for Bitcoin; model: A8, B3), B6 (is the VIX unbiased?; model: B5), B7 (Bitcoin premium; model: B5)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Black–Scholes and parity)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter12_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500 and VIX (1990–2026), VIX3M (2007–2026); daily closes up to 18 September 2026
- Bitcoin: daily closes; DVOL (Deribit, 2021–2026); the Bitcoin option chain of the lecture (Deribit snapshot)
- Two series together: prices joined on common days first, then returns

Formulas Used Today

- Parity: $C - P = Se^{-q\tau} - Ke^{-r\tau}$; binomial: $q = (R - d)/(u - d)$, $C = [qC_u + (1 - q)C_d]/R$ Cox, Ross & Rubinstein (1979)
- Black-Scholes Black & Scholes (1973): $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$
- Greeks: $\Delta = \Phi(d_1)$, $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, vega $= S\varphi(d_1)\sqrt{\tau}$, $\Theta = -S\varphi(d_1)\sigma/(2\sqrt{\tau}) - rKe^{-r\tau}\Phi(d_2)$
- Implied variance Cboe (2025): $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$
- SVI Gatheral & Jacquier (2014): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$, $w = \sigma_{imp}^2 \tau$, $k = \ln(K/F)$
- Variance risk premium Carr & Wu (2009): $VRP_t = IV_t^2 - RV_{t,t+h}$, $RV_{t,t+h} = \frac{A}{h} \sum_{i=1}^h r_{t+i}^2$ (A = periods per year)

A1: A Parity Arbitrage [Solved]

Task

- European options on a stock without dividends: $S = K = 50$, six months, $r = 4\%$
- Market quotes: call 4.20, put 2.80
- (a) Does parity hold? Which option is mispriced relative to the other?
- (b) Build the arbitrage and compute the riskless profit per share and per contract of 100 shares

Solution

- $Ke^{-r\tau} = 50e^{-0.02} = 49.01$; $S - Ke^{-r\tau} = 0.99$
- $C - P = 1.40 > 0.99$: the call is rich or the put is cheap; parity put price $= C - S + Ke^{-r\tau} = 3.21$
- Sell the call, buy the put, buy the share, borrow 49.01
- Cash today: $4.20 - 2.80 - 50 + 49.01 = 0.41$; at expiry:
 $-\max(S_T - 50, 0) + \max(50 - S_T, 0) + S_T - 50 = 0$
- Profit 0.41 per share, 40.99 per contract, before transaction costs

A2: Parity with a Dividend Yield [Proposed]

Solution discussed in class

Task

- Index options: $S = 80$, $K = 75$, three months, $r = 5\%$, dividend yield $q = 2\%$
- Market quotes: call 7.10, put 1.20
- (a) Check parity; (b) build the arbitrage; (c) what happens to your answer if you forget the dividend?
- Model: A1

A3: One-Period Binomial Tree [Solved]

Task

- $S = 100$; after one period $uS = 120$ or $dS = 90$; gross risk-free return $R = 1.05$
- Call with $K = 105$ expiring after one period [Cox, Ross & Rubinstein \(1979\)](#)
- (a) Replicating portfolio (Δ, B) and the call price
- (b) Risk-neutral probability q and the same price as an expectation

Solution

- $C_u = 15.00, C_d = 0; \Delta = \frac{15-0}{120-90} = 0.50$
- $B = \frac{uC_d - dC_u}{(u-d)R} = \frac{0 - 0.9 \cdot 15}{0.3 \cdot 1.05} = -42.86$
- $C = \Delta S + B = 50.00 - 42.86 \rightarrow$ buy half a share, borrow 42.86: $C = 7.14$
- $q = \frac{1.05 - 0.9}{1.2 - 0.9} = 0.50; C = \frac{0.5 \cdot 15 + 0.5 \cdot 0}{1.05} = 7.14$
- The real probability of the up move never appears

A4: An American Put on a Two-Period Tree [Proposed]

Solution discussed in class

Task

- Same $u = 1.2$, $d = 0.9$, $R = 1.05$ per period, $S = 100$; put with $K = 105$ expiring after two periods
- (a) Value the European put backwards through the tree
- (b) Value the American put: compare continuation and exercise at every node
- (c) Where is early exercise optimal, and why?
- Model: A3

A5: Black–Scholes and the Greeks, Step by Step [Solved]

Task

- $S = K = 100$, six months, $r = 3\%$, $\sigma = 25\%$, no dividends
- (a) d_1 , d_2 , the call and the put (via parity)
- (b) Delta, gamma, vega (per vol point) and theta (per calendar day) of the call
- (c) Approximate the call price if S jumps to 102 (delta-gamma)

Solution

- $\sigma\sqrt{\tau} = 0.1768$; $d_1 = \frac{0 + (0.03 + 0.03125) \cdot 0.5}{0.1768} = 0.1732$,
 $d_2 = -0.0035$
- $\Phi(d_1) = 0.5688$, $\Phi(d_2) = 0.4986$, $Ke^{-r\tau} = 98.51$:
 $C = 7.76$, $P = C - 100 + 98.51 = 6.27$
- $\Delta = 0.5688$; $\varphi(d_1) = 0.3930$, $\Gamma = \frac{0.3930}{100 \cdot 0.1768} = 0.0222$
- vega $= 100 \cdot 0.3930 \cdot \sqrt{0.5}/100 = 0.2779$; $\Theta = -0.0231$ per day
- $C(102) \approx 7.76 + 2\Delta + \frac{1}{2} \cdot 4 \Gamma = 8.942$; exact
 $C(102) = 8.942$

A6: A Delta-Gamma Hedge [Proposed]

Solution discussed in class

Task

- ▣ A dealer has **sold** 1 000 calls of A5 ($K = 100$)
- ▣ Available: a call with $K = 110$ (same expiry, price 3.90) and the stock
- ▣ (a) Delta and gamma of the dealer's position
- ▣ (b) Number of $K = 110$ calls that makes gamma 0, then number of shares that makes delta 0
- ▣ (c) Is the book now hedged against a change in volatility?
- ▣ Model: A5

A7: Implied Volatility by Newton's Method [Solved]

Task

- Call: $S = K = 100$, three months, $r = 2\%$; market price 4.50
- (a) Start from $\sigma_0 = 20\%$ and iterate
 $\sigma_{k+1} = \sigma_k - (BS(\sigma_k) - 4.50)/\text{vega}(\sigma_k)$
- (b) How many steps are needed for 4 decimals? Why so few?

Solution

k	σ_k (%)	$BS(\sigma_k)$	vega (per unit)
0	20.0000	4.2322	19.848
1	21.3495	4.5000	19.847
2	21.3495	4.5000	19.847

- $\sigma_1 = 0.20 - (4.2322 - 4.50)/19.848 = 21.3495\%$
- One step already gives 4 decimals (21.3495%): the ATM price is almost linear in σ , so Newton converges very fast
- Deep OTM options have a tiny vega: use a bracketing method (bisection, Brent) there

A8: Implied Variance from an Option Strip [Proposed]

Solution discussed in class

Task

- Forward $F = 100$, 90 days ($T = 90/365$), $r = 0$; OTM mid prices (puts below 100, calls above; at 100 the average of call and put):
- K : 80; 85; 90; 95; 100; 105; 110; 115; 120
 - ▶ $Q(K)$: 0.41; 0.60; 1.00; 1.87; 3.56; 1.43; 0.42; 0.09; 0.02
- (a) Apply the Cboe formula with $\Delta K = 5$ [Cboe \(2025\)](#); (b) compare 100σ with the ATM implied volatility (18.0%); (c) which strikes contribute most?
- Model: A7 (a table step by step) and the lecture example “A VIX for Bitcoin”

B1: Hedging Error and Rebalancing Frequency [Solved]

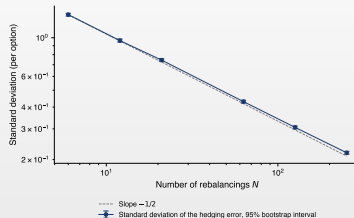
Task

- Sell a three-month ATM call ($S_0 = K = 100$, $\sigma = 20\%$, $r = 4\%$); 10 000 paths of geometric Brownian motion with $\mu = 8\%$
- (a) Delta-hedge with $N = 6, 12, 21, 63, 126, 252$ rebalancings; standard deviation of the hedging error
- (b) 95% bootstrap interval for each standard deviation (1 000 resamples of paths) [Efron \(1979\)](#)
- (c) Slope of $\ln(\text{s.d.})$ on $\ln N$ with a bootstrap interval

Interpretation

- Is the slope consistent with the theoretical $-1/2$?
- How often must a dealer rebalance to keep the error below 10% of the premium?
- What would transaction costs change?

B1: Results [Solved]



N	6	12	21	63	126	252
s.d.	1.356	0.963	0.744	0.430	0.306	0.219
95% CI	[1.331, 1.381]	[0.945, 0.983]	[0.731, 0.758]	[0.422, 0.438]	[0.300, 0.312]	[0.215, 0.223]

- Premium 4.49; slope -0.489 , 95% CI $[-0.496, -0.482]$: close to $-1/2$, slightly above it for these finite N
- Daily hedging: s.d. 9.6% of the premium: daily rebalancing is about enough for 10%; four times a day halves it again (ratio 1.97)
- Costs grow like N and the error falls like $1/\sqrt{N}$: there is an optimal, finite frequency

B2: Delta-Hedged S&P 500 Options, 1990–2026 [Proposed]

Solution discussed in class

Task

- Every 21 trading days sell a one-month ATM call on the S&P 500 at the VIX and hedge it daily at the same volatility (the lecture back-test) [Bakshi & Kapadia \(2003\)](#)
- (a) Mean monthly P&L with a Newey–West standard error (3 lags)
- (b) 95% interval by a block bootstrap (blocks of 6 months)
- (c) How much of the total gain did the five worst months take back?
- Model: B1 (bootstrap), B5 (Newey–West)

Interpretation

- Is the premium statistically significant?
- Why can a strategy with a high t -statistic still be dangerous?
- What do interest and dividends, ignored here, change?

B3: An SVI Smile for Bitcoin [Solved]

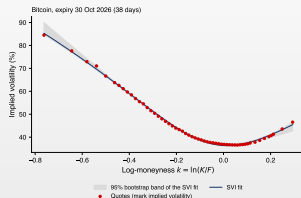
Task

- Deribit snapshot of the lecture; expiry closest to 30 days (2026-10-30, 38 days), 56 quoted OTM options
- (a) Fit SVI to the total variance $w(k)$ by least squares [Gatheral & Jacquier \(2014\)](#)
- (b) Pairs bootstrap (resample quotes, refit): 95% intervals for the ATM volatility and the skew $IV(0.15) - IV(-0.15)$
- (c) Check the no-butterfly condition $g(k) \geq 0$ of [Gatheral & Jacquier \(2014\)](#)

Interpretation

- Which parameter controls the skew, and what is its sign here?
- Is the skew significantly different from 0?
- Why do we fit w and not the prices?

B3: Results [Solved]



- ▣ $\hat{a} = -0.020$, $\hat{b} = 0.102$, $\hat{\rho} = -0.262$, $\hat{m} = -0.079$, $\hat{s} = 0.348$; root mean squared error 0.42 vol points
- ▣ ATM volatility 36.37%, 95% CI [36.10, 36.87]; skew -2.11 , 95% CI $[-2.73, -1.52]$: $\hat{\rho} < 0$ tilts the smile towards puts, and the interval excludes 0
- ▣ $\min_k g(k) = 0.136 > 0$: no butterfly arbitrage; the implied density is positive
- ▣ w is comparable across strikes and nearly linear in the wings; prices span four orders of magnitude

B4: A VIX for Bitcoin [Proposed]

Solution discussed in class

Task

- Same snapshot; the two expiries around 30 days; OTM options with a bid and an ask; USD mid price = BTC mid $\times$ forward; $r = 0$
- (a) Implied variance of each expiry with the Cboe formula [Cboe \(2025\)](#)
- (b) Interpolate to 30 days; compare with DVOL at the same moment [Alexander & Imeraj \(2020\)](#)
- Model: A8 (strip formula), B3 (the same snapshot)

Interpretation

- Why is the index above the ATM volatility of both expiries?
- Which choices (strike cut-off, mid vs mark prices) could explain the gap to DVOL?

B5: The Variance Risk Premium of the S&P 500 [Solved]

Task

- Daily, 1990–2026: $VRP_t = VIX_t^2 - RV_{t,t+21}$,
 $RV = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$, in %² Carr & Wu (2009)
- (a) Mean VRP with a Newey–West standard error (21 lags) and a 95% interval Newey & West (1987)
- (b) The same in vol points, $VIX_t - \sqrt{RV_{t,t+21}}$; share of positive days
- (c) Subperiods 1990–2007 and 2008–2026

Interpretation

- Why is the naive standard error wrong here, and by how much?
- Is the premium stable across subperiods?
- Why is the variance premium less precise than the vol-point premium?

B5: Results [Solved]

	n	Mean VRP (% ²)	95% CI	Vol points (s.e.)	Positive
1990–2026	9 198	113.0	[72.6, 153.4]	4.09 (0.27)	86%
1990–2007	4 513	149.0	[124.9, 173.2]	4.60 (0.25)	88%
2008–2026	4 685	78.3	[2.7, 153.9]	3.60 (0.46)	84%

- Naive s.e. 5.3 vs Newey–West 20.6: 3.9 times too small, because consecutive windows share 20 of 21 days
- The premium is positive in both subperiods; after 2008 the variance premium is barely significant ($t = 2.03$) because of 2008 and 2020
- Squaring magnifies the crisis windows: the variance premium has heavier tails than the vol-point premium

B6: Is the VIX an Unbiased Forecast? [Proposed]

Solution discussed in class

Task

- Regress $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ Mincer & Zarnowitz (1969)
- (a) Estimates with Newey–West standard errors (21 lags); joint Wald test of $\alpha = 0, \beta = 1$
- (b) Add the past realised variance $RV_{t-21,t}$: does it add information?
- Model: B5

Interpretation

- Is the VIX biased? In which direction?
- What does the regression with both variables say about market efficiency?

B7: The Variance Risk Premium of Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ DVOL (Deribit) 2021–2026 vs Bitcoin realised volatility over the next 30 calendar days, $A = 365$
- ▣ (a) Mean premium in vol points with a Newey–West standard error (30 lags); share of positive days
- ▣ (b) Compare with the S&P 500 in vol points and as the ratio implied / realised volatility
- ▣ Model: B5

Interpretation

- ▣ Is the Bitcoin premium larger or smaller than the equity premium? In which unit?
- ▣ How much can five years of data tell us?

C1: Is the Variance Risk Premium a Tradable Signal?

Question

- Would you sell variance on the S&P 500 or on Bitcoin every month, and when would you stop? Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- Instrument: a synthetic variance swap with vega notional 1, $P\&L = (IV^2 - RV)/(2 IV)$ vol points per month
- S&P 500 with the VIX (1990–2026); Bitcoin with DVOL (2021–2026)

What a good answer contains

- Sharpe ratio with a block-bootstrap interval, skewness, worst month, maximum drawdown
- A timing rule, e.g. skip months that start in backwardation (VIX above VIX3M); is it better out of sample?
- Comparison with the delta-hedged option of the lecture; position sizing and margin

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Price a 90-day European call on a non-dividend stock with $S = 100$, $K = 105$, volatility 20%, continuous rate $r = 4\%$; then give the put from put–call parity”
- The assistant answers with this code:

```
import numpy as np
from scipy.stats import norm
S, K, r, sigma = 100.0, 105.0, 0.04, 20          # sigma = 20%
T = 90 / 252                                     # 90 days
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put = call + S - K * np.exp(-r * T)              # put-call parity
print(f"call_{S}={call:.2f},_{K}put_{S}={put:.2f}")
```

- “The code follows Black–Scholes exactly; the put comes from parity, so no second formula is needed”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains **three** errors; everything else is correct
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a no-arbitrage bound, a unit check, the lecture, a second method
 - ▶ give the corrected line
- Run the corrected code; report C and P and check put–call parity numerically
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Before running the code: what should a 90-day call 5% out of the money cost, roughly: 0.2, 2 or 20?
- Which error would still give a plausible-looking price? Which check catches it?
- Why is “the put comes from parity” not a check of the call price?

Common Pitfalls

- ▣ Forgetting dividends or the interest rate in parity: a false arbitrage
- ▣ Treating the risk-neutral probability as a forecast
- ▣ Quoting vega or theta without the unit (per vol point? per day? per year?)
- ▣ Newton's method on deep OTM options ($\text{vega} \approx 0$): use a bracketing solver
- ▣ Naive standard errors on overlapping windows: t -statistics several times too large
- ▣ Reading a high Sharpe ratio of a short-volatility strategy without its skewness and worst month

Take-Home Assignment (Team Project)

- ▣ Take a new Deribit snapshot for Ether (ETH) options and build the smile of three expiries with SVI, with bootstrap intervals
- ▣ Compute a 30-day VIX-type index and the risk-neutral density; compare with Bitcoin [Pele et al. \(2023\)](#)
- ▣ Delta-hedge a sold one-week ATM option on historical Bitcoin paths; report the error with a block-bootstrap interval [Matic, Packham & Härdle \(2023\)](#)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

-  XFG Advanced Methods in Quantitative Finance
-  Pricing Kernels and Risk Premia
-  Hedging Cryptocurrency Options

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

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